

1. 1024 samples per transducer is recorded in the memory.

$$1024 \times 64 = 65,536$$

64 element array is
divided into $= \frac{64}{4} = 16$

The memory can be accessed as:

$$\text{Group } g \rightarrow \{4g, 4g+1, 4g+2, 4g+3\}$$

$$g = \{0, \dots, 15\}$$

2. TDM controller starts processing the 16 groups $\Rightarrow$ Always hardware reuse for all datapaths.

3. Our architecture does not calculate new geometric delays for every image pixel

Instead we have selected four fixed focal depths:

$$z_f = \{5, 9, 19, 29\} \text{ mm}$$

Controller selects one focus at a time:

$$\text{focus_sel} = \{0, 1, 2, 3\}$$

$\rightarrow$ For specific focal zones, specific delays are obtained & applied to four ADC channels.

4. In the delay LUT, the precomputed delays are indexed by (g, i)

where 'i' is the channel index of each transducer present inside the 16 groups.

For each $g \in \{0, 1, 2, \dots, 15\}$



D_1, D_2, D_3, D_4

$$D_i = (D_{int})_i + \frac{(D_{frac})_i}{2^8}$$

→ These values are precomputed using a python script.

5. Delay address generation

If current ptr is $\rightarrow x[n_i]$

we will read previous samples according to delays.

$$\therefore x[n_i - D_i] \text{ \& } x[(n_i - D_i) + 1]$$

→ Then according to fractional delay value, we generate an interpolated value with delayed index & its precursor.

$$(x_{interpolated})_i = \alpha_i x[(n_i - D_i) + 1] + (1 - \alpha_i) x[n_i]$$

$$\alpha_i = \frac{(D_{\text{trace}})_i}{256}$$

Within a single group, $i = \{1, 2, 3, 4\}$

Hence, we get $(\alpha_{\text{interpolated}})_1, (\alpha_{\text{interpolated}})_2, \dots, (\alpha_{\text{interpolated}})_3, (\alpha_{\text{interpolated}})_4$.

$$\therefore (DAS)_{\text{group} = \text{constant}} = \sum_{i=1}^4 (\alpha_{\text{interpolated}})_i$$

★ We are not summing 64 elements into one value

→ 16 groups are processed & sequentially.

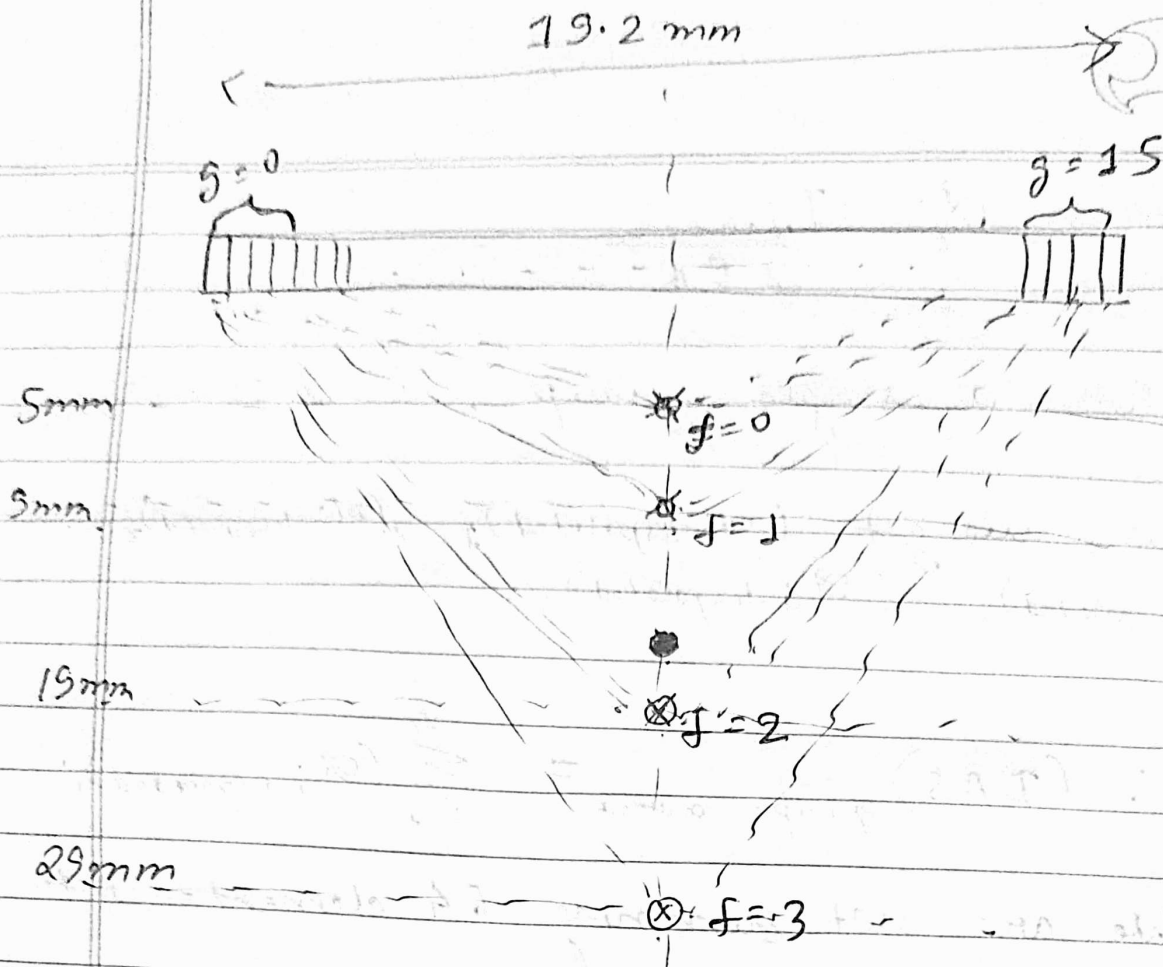
6. Process is repeated for all depth samples for a particular group & focus.

16 groups $\times$ 4 foci $\times$ 1024 samples

= 65,536 outputs per frame

Each output can be indexed as: $DAS[g, f, i]$

Note: We trace each group for a particular focal zone at once, then we move onto other focal zones.



$$\text{pitch} = 0.3 \text{ mm}$$

$$\text{total aperture} = 19.2 \text{ mm} = 64 \times 0.3 \text{ mm}$$

$$\text{element } (i^{\text{th}}) \text{ coordinate} = (i - 31.5) \times 0.3 \text{ mm}$$

$$C_{\text{sound}} = 1540 \text{ m/s}$$

$$f_s = 20 \text{ MHz}$$

$$\text{Depth resolution} = \frac{(C_{\text{sound}} \times T_s)}{2} = 0.0385 \text{ mm/sample}$$

$$\text{Acquisition depth} = 1024 \text{ samples} \times 0.0385 \\ = 39.4 \text{ mm}$$

$$\text{ADC input format} = 11 \text{ bit signed data} \\ (-512 \text{ to } +511)$$